src\LDR_LED\LDR_LED.ino

```
1  // LDR_LED.ino
2  int fromLow = 100;
3  int fromHigh = 400;
4  // Note: toLow and toHigh are inverted to brighten LED when ldrValue increases
5  int toLow = 255;
6  int toHigh = 0;
7  int ldrValue; // LDR value as read by the analog input
8  int ledBrightness;  // mapped LED brightness (0-255)
9
10 void setup() {
11     // Initialize serial communication
12     Serial.begin(9600);
13     Serial.println("LDR Value, LED Brightness");
14 }
15
16 void loop() {
17     // Read the LDR value
18     ldrValue = analogRead(A0);
19
20     // Map the LDR value to LED brightness (0-255)
21     // Increase the value of fromLow for a more pronounced result
22     int ledBrightness = map(ldrValue, fromLow, fromHigh, toLow, toHigh);
23
24     // Set the LED brightness
25     analogWrite(9, ledBrightness);
26
27     // Print the LDR value and LED brightness to the serial monitor
28     Serial.print(ldrValue);
29     Serial.print(", ");
30     Serial.println(ledBrightness);
31
32     delay(100);
33 }
34
```